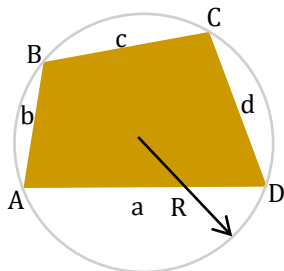




7. Cyclic quadrilateral



$$A = \sqrt{(s-a)(s-b)(s-c)(s-d)}$$

Where:

$$s = \frac{a+b+c+d}{2}$$

Radius of circumcenter, R:

$$R = \frac{\sqrt{(ab+cd)(ac+bd)(ad+bc)}}{4A}$$

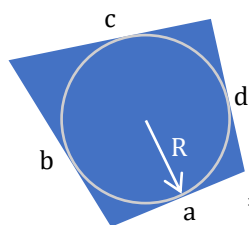
Where: A = area of quadrilateral

$$R = \frac{1}{4} \sqrt{\frac{(ab+cd)(ac+bd)(ad+bc)}{(s-a)(s-b)(s-c)(s-d)}}$$



Above is known as Parameshvara's circumradius formula. It was derived in the 15th century by the Indian mathematician Vatasseri Parameshvara.

8. Quadrilateral circumscribing a circle



$$A = Rs$$

$$A = \sqrt{abcd}$$

Where:

$$s = \frac{a+b+c+d}{2}$$

Brahmagupta's Theorem

"In a cyclic quadrilateral having perpendicular diagonals, the perpendicular to a side from the point of intersection of the diagonals always bisects the opposite side."

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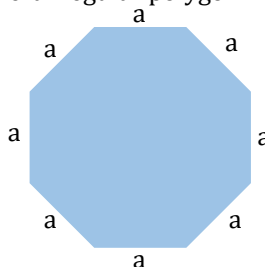
Brahmagupta (598-665 AD) was a Hindu astronomer and mathematician who became head of the observatory at Ujjain – the foremost mathematical center in ancient India.

Ptolemy's Theorem

"The sum of the products of two pairs of opposite sides of a convex cyclic quadrilateral is equal to the product of the lengths of the diagonals."

Regular polygons

1. General regular polygon

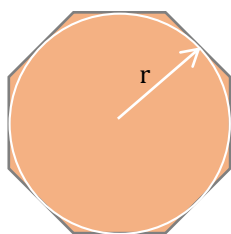


$$\text{Area} = \frac{1}{4} na^2 \cot\left(\frac{180^\circ}{n}\right)$$

$$\text{Perimeter} = na$$

Where: n = number of sides

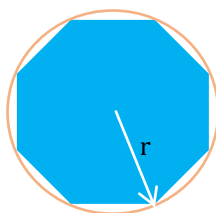
2. Regular polygon circumscribing a circle



$$\text{Area} = nr^2 \tan\left(\frac{180^\circ}{n}\right)$$

$$\text{Perimeter} = 2nr \tan\left(\frac{180^\circ}{n}\right)$$

3. Regular polygon inscribed in a circle

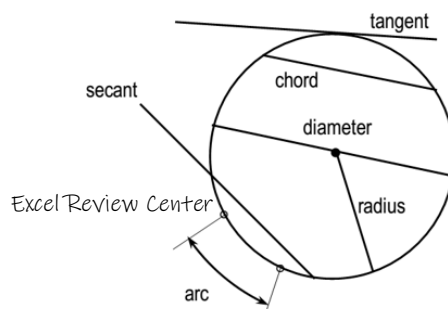


$$\text{Area} = \frac{1}{2} nr^2 \sin\left(\frac{360^\circ}{n}\right)$$

$$\text{Perimeter} = 2nr \sin\left(\frac{180^\circ}{n}\right)$$

Circles

"A circle is a plane curve that is the locus of all points in the plane equidistant from a given point, called center. The term "circle" comes from the Latin "circus", which refers to a large round or rounded oblong enclosure in which the famous Roman chariot races were held. The circle is a conic section whose eccentricity is zero.

Terms related to a circle

1. **Arc** – length of circle between two points on the circle or between two radii.
2. **Tangent to a circle** – a line touching the circle in one place. A tangent is perpendicular to the radius of the circle.
3. **Secant of a circle** – a line cutting the circle in two places.
4. **Diameter** – the longest chord of a circle that passes through the center.
5. **Radius** – the distance from the center to the circle. It is one-half of the diameter.
6. **Chord** – the segment of a secant bounded by the circle.
7. **Circumference** – the perimeter of a circle. This is also known as periphery. The circumference of a circle is $2\pi(\text{radius})$ or $\pi(\text{diameter})$.
8. **Radian** – the measure of an angle whose arc length is equal to the radius of the circle.
9. **Sector of a circle** – area bounded by two radii and the included arc.
10. **Segment of a circle** – area bounded by a chord and the arc subtending the chord.

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